Soil Classification — Quick Notes

A revision summary for the Soil Mechanics course on this platform

Why classify soil at all

Two soils that look similar can behave very differently under a foundation. Classification groups soils by properties that actually predict engineering behaviour — grain size and plasticity — so an engineer can anticipate strength, settlement, and drainage behaviour before design even starts.

Grain size groups

- **Gravel** — particles larger than 4.75 mm (retained on a No. 4 sieve)
- **Sand** — between 4.75 mm and 0.075 mm (No. 4 to No. 200 sieve)
- **Silt** — finer than 0.075 mm, low plasticity, particles still roughly bulky
- **Clay** — finer than 0.075 mm, plate-shaped particles, plastic when wet

Atterberg limits, in one line each

- **Liquid Limit (LL)** — moisture content where soil starts to flow like a liquid
- **Plastic Limit (PL)** — moisture content where soil starts to crumble instead of roll into a thread
- **Plasticity Index (PI = LL − PL)** — the moisture range over which the soil stays plastic; a wider range means more clay-like behaviour

Unified Soil Classification System (USCS) — group symbols

Symbol	Meaning	Typical behaviour
GW / GP	Well / poorly graded gravel	Good drainage, good bearing, low compressibility
SW / SP	Well / poorly graded sand	Good drainage, moderate bearing
ML / CL	Low-plasticity silt / clay	Moderate compressibility, moisture-sensitive
MH / CH	High-plasticity silt / clay	High compressibility, significant settlement risk

Quick recall

Coarse-grained soils (gravel, sand) are classified mainly by grain size and gradation. Fine-grained soils (silt, clay) are classified mainly by plasticity — which is exactly what the Atterberg Limits lab on this platform measures. Try the lab after reading this to see LL, PL, and PI calculated from simulated test data.